A Comparative Study of Bacterial Contamination and Growth on Frequently Touched Dry Surfaces in Public Spaces

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Introduction

The research titled Bacterial Growth on Dry Surfaces In Public Spaces aims to research the growth of bacteria in a school environment. This study was conducted at Helios Middle School in Sunnyvale, California. Data for this experiment was collected in two locations: an outside door handle which students use to travel from the courtyard lunch area to the 7-8 grade humanities room, and an outdoor stairway railing which students use to walk up from the courtyard lunch area to the 7-8 grade science room. This study relied on the underlying principles of germ theory and bacterial transmission. Germ theory, invented by scientists Louis Pasteur and Robert Koch, explained that microbes cause diseases. In the case of bacteria, it can infect a human via contact with an opening on the skin or through the air inhaled through the nose or mouth (Council 2026). Bacterial transmission is the principle that bacteria can infect other organisms if they come in contact. Bacteria achieve this by propagating via a frequently touched surface or by being transmitted through the air via water droplets and air particles (Doron et al. 2008). This study focused on bacterial transmissions via frequently touched surfaces at a middle school, specifically the handle and railing mentioned above. The next section, Hypothesis, will further explain the setup for this experiment and the hypothesis it followed.

Hypothesis

The locations chosen to study included a door handle and a stairway railing in a school courtyard. The independent variable of this study was location of sample, and the dependent variable was bacterial growth amount. The hypothesis of this experiment was that the door handle will have a greater amount of bacterial growth because hands from Helios students touch it hundreds of times every day to get from the courtyard to the 7-8 humanities room. However, the railing will have less bacterial growth because it was observed that the frequency of students using the railing is lower than that of the handle. Furthermore, the points of contact for someone's hand to be placed on the railing far outnumber the points of contact on the doorhandle by several hundreds, increasing the concentration of bacteria on the smaller door handle. Bacteria can transmit via hand contact from one place to another, due to the fact that our hands are excellent places for bacteria to survive and that our hands touch lots of surfaces each day. Therefore, I hypothesized that the sample with more hand contact (the handle) will have more bacteria than the one with less hand contact (railing).

Methods and Materials

The experiment itself consisted of swabbing two locations (explained above under Hypothesis) with single-ended cotton swabs. The experiment was performed in the afternoon on a sunny Monday with typical Californian Mediterranean climate for the month of May. The independent variable of this experiment was swabbing location, and the dependent variable was bacterial growth amount. Both surfaces were metallic, the railing painted green and the handle worn down to a golden color (from hand surface oils) commonly found on frequently touched metals. The swabbing was done by rotating the cotton head of each swab across each surface, ensuring that any bacterial contamination was collected. From there, the cotton head's contents were directly transferred to an open agar plate, then closed with a Petri dish. Finally, the Petri dish was placed in a 37 degree (Celsius) incubator for 3 days and all other materials were cleaned or disposed of to avoid further contamination of the sample. Our experiment controlled variables including swabbing method, general vicinity of samplings, a clean/sterile/uncontaminated Petri dish, method of bacterial growth, and time of day/climate. The Petri dish was measured after 3 days in the incubator by calculating the estimated bacterial spread (cover) of four quadrants drawn on the lid of the sealed dish, then taking the arithmetic mean of those four percentages.

Results

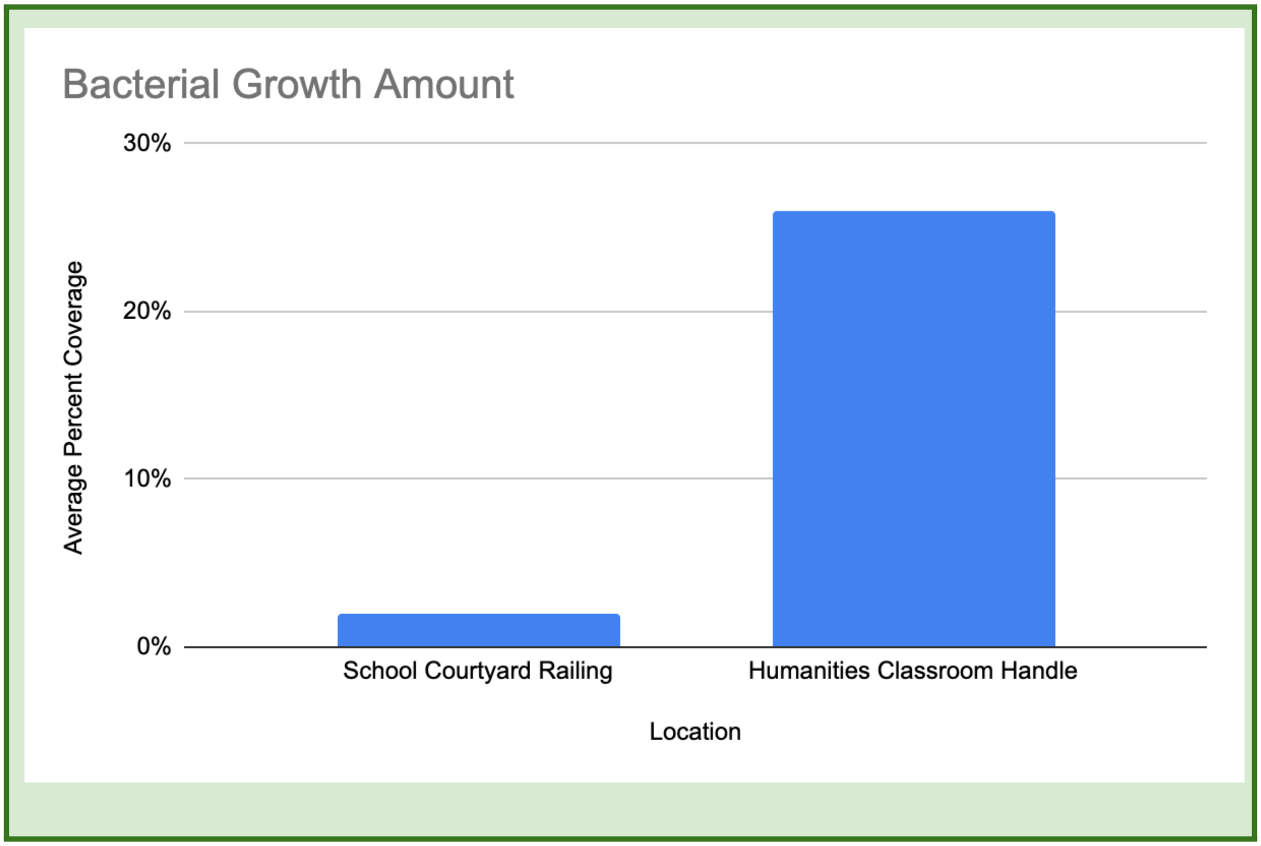

Fig.1: "Bacterial Growth Amount"

As shown in Fig. 1, the average bacterial growth from the Middle School Courtyard Stairway Railing was only 2%. However, the average bacterial growth from the handle on the way to classroom was 26%, 24% higher than the railing. Again, the swabbing method was nearly identical in both locations, all samples were taken roughly at the same time of day, and both petri dishes underwent the same growth process.

Discussion

Overall, the handle had a far greater concentration of bacteria than the railing- a difference of 24%. The hypothesis was supported by this difference between samples, and for the reasons mentioned earlier, frequency and points of contact. This finding supports our hypothesis's underlying assumption that a handle the length of a few inches touched hundreds of times per day by human hands would be expected to have more bacteria than a meter-long railing touched a few times per day. Similar studies have found the same result, such as a door handle swabbing study done at Port Harcourt's university hospital in Nigeria (Edi et al. 2023). This study found that door handles touched with higher frequencies in a hospital, such as in a ward or clinic room, were far more concentrated with bacteria than door handles touched with lower frequencies, such as in an office or laboratory. A slightly more surprising finding was that the lowest concentration of bacteria was found in the public toilets, a frequently used space. This surprising finding raises questions for future study in the Port Harcourt University Hospital, including the type of bacteria present and the frequency of patients/staff washing hands (Edi et al. 2023).

Despite its impactful results, our study has an opportunity for further refinements. Uncontrolled variables that could have impacted this study included accuracy of estimation in measurement and a possible currently undetermined third variable (i.e. if the door handles/railings were recently cleaned, if the presence of students in the school was greater/less than average daily attendance) other than frequency or spread that could have influenced results. Additionally, the type of bacteria collected was not measured, and therefore could have created misleading results (e.g. only one type of bacteria was on the handle while hundreds of species were on the railing, indicating higher bacterial contact, but it was not measured in the Petri dish. Another possibility would be that a different handle material/shape could indicate different bacterial growth rates.) Furthermore, the railing was cracked painted metal and the handle was smooth metal, which could have impacted bacterial growth slightly. Future experiments conducted on bacterial spread through hand contact should account for these uncontrolled variables by using more precise measurements of results and more advanced scientific measurement methods. In addition, the similar study conducted in Nigeria's Port Harcourt University Hospital (Edi et al. 2023) had an issue with the time period of regular hospital cleanings, a third variable that could have impacted their study results (e.g. different samples coming from doorhandles cleaned at different times).

Because bacteria can transmit via hand contact, the surface with more concentrated hand contact, such as the handle, should have more bacteria present than a surface with less concentrated hand contact, such as the railing. This is previously researched knowledge about bacteria (Cobrado et al. 2017) that is consistent with the results of this experiment. According to a meta analysis study conducted on microbial contamination of door handles, "These findings emphasise the need for targeted hygiene interventions to reduce the risk of pathogen transmission via door handles, particularly in high-traffic areas (Appiah et al. 2025)." In other words, while bacterial transmission via public surfaces like door handles seems like a nuisance, it is actually a pressing issue for the various laboratories dedicated to understanding bacterial safety. Understanding how to keep the public safe from pathogenic bacterial transmission is a key part of understanding how bacteria behave and affect humans everywhere at schools, hospitals, and other public spaces.

Conclusion

The hypothesis of this experimental study was a prediction that the handle would have a greater concentration of bacteria due to more frequent and compacted hand contact. The hypothesis was confirmed as explained in the results section. The 24% (26%-2%) difference of the 7-8 Humanities handle and School Courtyard railing samples explained that it matters what surfaces are frequently touched in public spaces, not what the public perception of a dirty surface is. An example of this is the handle utilized in this experiment- it is touched daily without frequent thought from people passing by. Future iterations of this experiment should definitely have more precise measurements, such as via a microscope or a computer scanner. Overall, this experiment raised questions about how bacterial contamination is frequently overlooked as a public issue, like that of a door handle (Simeonova 2024). It is important to prioritize understanding bacteria through experiments like these to keep the public safe from pathogenic bacterial growth in public spaces, especially schools. Further studies on bacterial growth in other public spaces like malls, libraries, restaurants, and transportation will deepen the scientific community's understanding of bacteria and allow scientists to keep the public informed about pathogenic bacteria in a public environment.

References/Appendix

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